25900896 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025 FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCMBG100 - UNSEEN WORLD OF MICROBES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** out of **12** Questions

Each Question carries **2** marks

1. Mention any two scientists who contributed to Golden era of Microbiology. [K] / [CO1]
2. What is mRNA therapy? [K] / [CO2]
3. Write a notes on Protozoa? [U] / [CO3]
4. Mention the characteristics of organisms classified under kingdom animalia? [K] / [CO2]
5. What are some examples of beneficial microbes found in the human body, and what roles do they play? [U] / [CO3]
6. Name the contribution of Francesco Redi [K] / [CO1]
7. What was the first antibiotic discovered? [K] / [CO1]
8. How does the domain Bacteria differ from Archaea? [K] / [CO2]
9. What is mRNA therapy? [K] / [CO2]

10. What is monochrome staining [U] / [CO3]

11. Give an example for an acidic stain and basic stain [U] / [CO3]

12. Cite the roles of a medical microbiologist [K] / [CO1]

[10x2 = 20]

Part B

Short Essay Questions

Answer any **4** out of **6** Questions

Each Question carries **5** marks

13. Write a note on Linnaeu's 2 kingdom classification? [K] / [CO2]

14. List five structural differences between prokaryotes and eukaryotes. [U] / [CO3]

15. Write a notes on Archaebacteria ? [U] / [CO3]

16. Identify the beneficial and harmful role of microorganisms in food and dairy industry. [K] / [CO1]

17. Describe the biological sorces of Antimicrobial peptides. [U] / [CO2]

18. Relate medical microbiology to public health. [K] / [CO1]

[4x5 = 20]

Part C

Essay Type Questions

Answer any **1** out of 2 Questions

Each Question carries **10** marks

19. State Biogenesis vs Abiogenesis. Mention the experiments supporting both the theories. [K] / [CO1]

20. What are the main characteristics and features of Viruses. [U] / [CO3]

[1x10 = 10]